

**UNIVERSITY OF THE FREE STATE  
UNIVERSITEIT VAN DIE VRYSTAAT**

BLOEMFONTEIN CAMPUS/KAMPUS

**CSIS3744**

**DEPARTMENT: COMPUTER SCIENCE AND INFORMATICS  
DEPARTEMENT: REKENAARWETENSKAP EN INFORMATIKA**

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**EXAMINATION/EKSAMEN:  
Main End-of-year/Hoofjaareind 2017  
PAPER/VRAESTEL 1**

**ASSESSOR(S)/  
ASSESSOR(E): 1. Mnr. W.S.J. Marais**

**MODERATOR(S)/  
MODERATOR(E): 1. Dr. B.P. Haskins**

**TIME/TYD: 180 min MARKS/PUNTE: 100**

**INSTRUCTIONS:** Answer all the questions. Number your questions clearly and correctly. The Afrikaans versions of the questions are in italics.

**INSTRUKSIES:** *Beantwoord al die vrae. Nommer u antwoorde duidelik en korrek. Die Afrikaanse weergawe van die vrae is kursief gedruk.*

**Question / Vraag 1 [13 Marks / Punte]**

Netcom wants to install communication network infrastructure between certain cities as specified next: Johannesburg, Pretoria, Durban, East London, Port Elizabeth and Cape Town have to form a physical ring topology by means of a SONET network. Each of these specified cities also has to be connected by means of T-4 lines in a physical star topology with Bloemfontein as the central WAN site. Furthermore, the SONET ring has to be self-healing. /

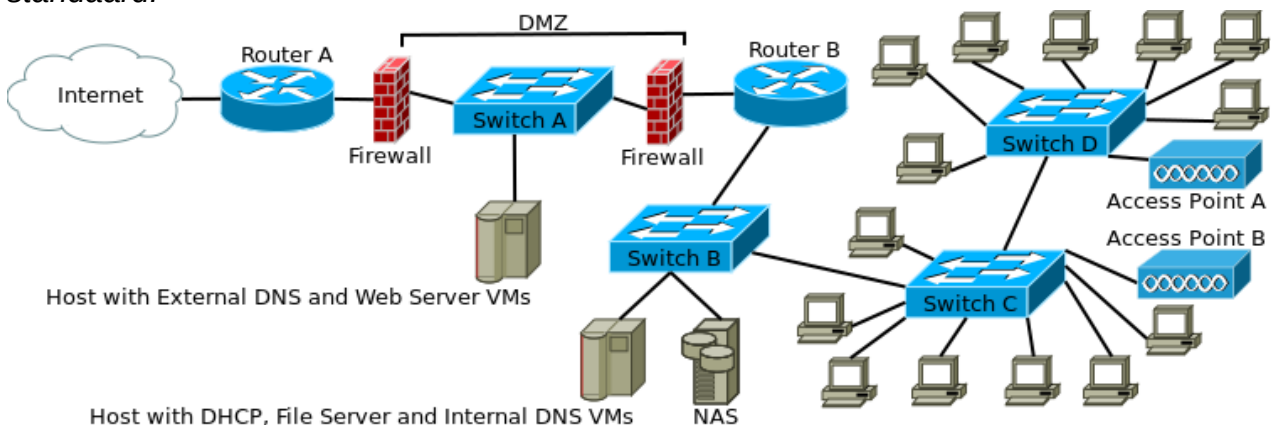
*Netcom wil kommunikasienetwerk-infrastruktuur tussen sekere stede, soos hierna gespesifiseer, installeer: Johannesburg, Pretoria, Durban, Oos-Londen, Port Elizabeth en Kaapstad moet 'n fisiese ringtopologie met behulp van 'n SONET-netwerk vorm. Elke een van hierdie gespesifiseerde stede moet ook met behulp van T-4 lyne in 'n fisiese stertopologie met Bloemfontein as die sentrale WAN-werf verbind word. Verder nog, moet die SONET-ring selfgenesend wees.*

- 1.1 Draw a diagram to illustrate the physical topology of Netcom's WAN. /  
*Teken 'n diagram om die fisiese topologie van Netcom se WAN te illustreer.* (5)

- 1.2 Discuss the greatest drawback of a star topology WAN and explain how Netcom's topology will circumvent this drawback. /  
*Bespreek die grootste nadeel van 'n ster-topologie-WAN en verduidelik hoe Netcom se topologie hierdie nadeel sal omseil.* (2)
- 1.3 What type of signalling will be used on Netcom's network backbone? /  
*Watter tipe seinversending sal gebruik word op Netcom se netwerkruggraat?* (1)
- 1.4 What type of multiplexing is used on a SONET network? /  
*Watter tipe multipleksering word op 'n SONET-netwerk gebruik?* (1)
- 1.4.1 Explain how this type of multiplexing is executed. /  
*Verduidelik hoe hierdie tipe multipleksering uitgevoer word.* (4)
- [13]

### **Question / Vraag 2 [30 Marks / Punte]**

The following is a network diagram of a client/server LAN that belongs to MX Enterprises. The network backbone implements the 1000Base-SX Ethernet standard: /  
*Die volgende is 'n netwerkdiagram van 'n kliënt/bediener-LAN wat aan MX-Enterprises behoort. Die netwerkruggraat implementeer die 1000Base-SX Ethernet-standaard:*



- 2.1 Explain the meaning of the Ethernet standard that has been implemented on the network backbone. /  
*Verduidelik die betekenis van die Ethernet-standaard wat op die netwerkruggraat geïmplementeer is.* (3)
- 2.1.1 Which physical media conforms to the specified Ethernet standard? /  
*Watter fisiese media voldoen aan die gespesifiseerde Ethernet-standaard?* (1)
- 2.1.2 What is the maximum segment length for the specified Ethernet standard? /  
*Wat is die maksimum segmentlengte vir die gespesifiseerde Ethernet-standaard?* (1)
- 2.1.3 Briefly discuss two factors that influence the maximum segment length of the specified Ethernet standard.  
*Bespreek kortliks twee faktore wat die maksimum segmentlengte beïnvloed van die gespesifiseerde Ethernet-standaard.* (2)

- 2.2 Specify how many collision domains and how many broadcast domains the current LAN contains. /  
*Spesifiseer hoeveel botsing-domeine en hoeveel uitsaai-domeine die huidige LAN bevat.* (2)
- 2.2.1 Explain the difference between a collision and a broadcast domain. /  
*Verduidelik die verskil tussen 'n botsing- en 'n uitsaai-domein.* (2)
- 2.3 Explain the purpose of the demilitarised zone (DMZ) with regard to the DNS and web servers. /  
*Verduidelik die doel van die gedemilitariseerde sone (DMZ) ten opsigte van die DNS- en webbedieners.* (3)
- 2.4 Specify the router type (Interior, exterior or border) for router A and router B. /  
*Spesifiseer die tipe roetebepaler (Interne, eksterne of grens) vir roetebepaler-A en roetebepaler-B.* (2)
- 2.4.1 Discuss the similarities and differences between the IS-IS and OSPF routing protocols. /  
*Bespreek die ooreenkomste en verskille tussen die IS-IS- en OSPF-roetebepalingsprotokolle.* (4)
- 2.4.2 Which of the two routers (A and/or B) will implement the IS-IS routing protocol? /  
*Watter van die twee roetebepalers (A en/of B) sal die IS-IS-roetebepalingsprotokol implementeer?* (1)
- 2.5 What is the purpose of the NAS device? /  
*Wat is die doel van die NAS-toestel?* (1)
- 2.5.1 Discuss the advantages of having a NAS over a typical file server. /  
*Bespreek die voordele van 'n NAS oor 'n tipiese lêerbediener.* (3)
- 2.5.2 Why is the file server then still necessary in spite of the network containing a NAS? /  
*Waarom is de lêer-bediener dan nogsteeds nodig, ten spyte daarvan dat die netwerk 'n NAS bevat?* (1)
- 2.6 Explain the difference between baseband and broadband transmissions and specify where these transmission types are used on MX Enterprises' network? /  
*Verduidelik die verskil tussen basisband- en breëband-uitsendings en spesifiseer waar hierdie tipe uitsendings op MX-Enterprises se netwerk gebruik word?* (4)

[30]

### **Question / Vraag 3 [15 Marks / Punte]**

MX Enterprises' wireless access point A and B are configured as follows: /  
 MX-Enterprises se draadlose toegangspunt-A en -B is as volg opgestel:

- Access point A's SSID: MXAWiFi / Toegangspunt A se SSID: MXAWiFi
- Access point B's SSID: MXBWiFi / Toegangspunt B se SSID: MXBWiFi
- Encryption standard: WPA2 / Enkripsiestandaard: WPA2
- DHCP: Enabled / DHCP: Aangeskakel
- Wi-Fi Protected Setup: Enabled / WiFi Beskermd Opstelling: Aangeskakel
- Supported 802.11 standards: n/ac / Ondersteunde 802.11 standaarde: n/ac

- 3.1 Identify and explain all the relevant security risks that are contained in the access point's configuration settings and give a solution for the vulnerabilities. / *Identifiseer en verduidelik al die tersaaklike veiligheidsrisiko's vervat in die toegangspunt se konfigurasie-instellings en gee 'n oplossing vir die kwesbaarhede.* (2)
- 3.2 Explain why WPA2 is a better encryption protocol than WEP. / *Verduidelik waarom WPA2 'n beter enkripsie-protokol is as WEP.* (2)
- 3.3 Name and discuss the access method that is implemented by the wireless access points. / *Noem en bespreek die toegangsmetode wat deur die draadlose toegangspunte geïmplementeer word.* (4)
- 3.4 Does the wireless network make use of simplex, half-duplex or full-duplex transmission? / *Maak die draadlose netwerk gebruik van simpleks, half-dupleks of vol-dupleks transmissie?* (1)
- 3.5 Explain how the wireless network can be set up to allow the users to roam between the access points. / *Verduidelik hoe die draadlose netwerk opgestel kan word om die gebruikers toe te laat om rond te beweeg tussen die toegangspunte.* (2)
- 3.6 Explain the difference between passive scanning and active scanning during the association process. / *Veduidelik die verskil tussen pasiewe skandering en aktiewe skandering gedurende die assosiasie proses.* (2)
- 3.7 On what levels of the OSI model do the following devices operate? / *Op watter vlakke van die OSI-model funksioneer die volgende toestelle?*
- 3.7.1 Wi-Fi access point. / *Wi-Fi toegangspunt.* (1)
- 3.7.2 Router. / *Roetebepaler.* (1)
- [15]

#### **Question / Vraag 4 [16 Marks / Punte]**

MX Enterprises wants to separate their servers on the internal network from the workstations and access points by means of 2 VLANs. /

*MX-Enterprises wil hul bedieners op die interne netwerk skei van die werkstasies en toegangspunte deur middel van 2 VLAN's.*

- 4.1 Which connectivity device would the administrator have to reconfigure to implement the VLANs? / *Watter verbindingstoestel sal die administrateur moet verstel om die VLAN's te implementeer?* (1)
- 4.2 What is the purpose of an access port and a trunk port on a switch respectively? / *Wat is die doel van 'n toegangspoort en 'n stampoort op 'n skakelaar onderskeidelik?* (2)

- 4.3 Explain the difference between how a switch and hub operates. /  
Verduidelik die verskil tussen hoe 'n skakelaar en spil funksioneer. (2)
- 4.4 MX Enterprises wants to create a subnet for each VLAN. Their internal network address is 192.168.56.0. /  
MX-Enterprises wil 'n subnetwerk vir elke VLAN skep. Hul interne netwerkadres is 192.168.56.0.
- 4.4.1 Calculate the new subnet mask for the proposed subnets. /  
Bereken die nuwe subnetmasker vir die voorgestelde subnetwerke. (4)
- 4.4.2 How many usable addresses are available per subnet? /  
Hoeveel bruikbare adresse is beskikbaar per subnetwerk? (2)
- 4.4.3 How many machines will the whole network be able to host? /  
Hoeveel masjiene sal die hele netwerk kan akkomodeer? (1)
- 4.5 Complete the table below by supplying the network address, broadcast address and usable host address range for the 2 subnets: /  
Voltooi die onderstaande tabel met die netwerkadres, uitsaai-adres en bruikbare gasheer-adresreeks vir die 2 subnetwerke:

|       | Subnet   | Network Address<br>Netwerkadres | Broadcast Address<br>Uitsaai-adres | Usable Host Address Range<br>Bruikbare Gasheer-adresreeks |
|-------|----------|---------------------------------|------------------------------------|-----------------------------------------------------------|
| 4.5.1 | <b>1</b> |                                 |                                    |                                                           |
| 4.5.2 | <b>2</b> |                                 |                                    |                                                           |

(2)

(2)

[16]

### **Question / Vraag 5 [26 Marks / Punte]**

The domain name of MX Enterprises is "mx.co.za". The following table displays the host names and IPv4 or IPv6 addresses of the virtual servers that are hosted on two separate machines connected to switch A and B respectively: /

Die domeinnaam van MX-Enterprises is "mx.co.za". In die volgende tabel word die gasheername en IPv4- of IPv6-adresse van die virtuele bedieners wat op twee afsonderlike masjiene gehuisves word, gekoppel aan skakelaar A en B onderskeidelik:

| Nr | Network Device<br>Netwerkttoestel | Host Name<br>Gasheernaam | IP Address<br>IP-Adres                  |
|----|-----------------------------------|--------------------------|-----------------------------------------|
| a) | Web Server /<br>Webbediener       | web                      | 146.182.12.18                           |
| b) | DNS Server 1 /<br>DNS-bediener 1  | dns1                     | 146.182.18.18                           |
| c) | DHCP Server /<br>DHCP-bediener    | dhcp                     | fe80:0000:0000:0000:e863:ac78:0b36:7e2d |
| d) | File Server /<br>Lêerbediener     | file                     | 2002:ac56:00de:0000:a8dd:31ed:769a:63e7 |
| e) | DNS Server 2 /<br>DNS-bediener 2  | dns2                     | fe80:0000:0000:0000:ea61:ac78:0b3a:23cd |

**Table 1 / Tabel 1**

Answer the following questions with regard to Table 1 on page 5. Indicate in your answers which network device or IP address you are referring to by means of the device's host name. /

*Beantwoord die volgende vrae met betrekking tot Tabel 1 op bladsy 5. Dui in jou antwoord aan na watter netwerktoestel of IP-adres jy verwys deur middel van die toestel se gasheernaam.*

- 5.1 Specify the IPv6 address type for the DHCP and file server. /  
*Spesifiseer die IPv6-adrestipe vir die DHCP- en lêerbediener.* (2)
- 5.2 Specify the IPv4 class that the web server's IP address belongs to. /  
*Spesifiseer die IPv4-klas waaraan die webbediener se IP-adres behoort.* (1)
- 5.3 What is the default subnet mask for DNS server 1's (dns1) IP address? /  
*Wat is die verstek subnetmasker vir die DNS-bediener 1 (dns1) se IPv4-adres?* (1)
- 5.4 Provide the interface ID's for each of the DNS servers. /  
*Verskaf die koppelvlak-ID's vir elk van die DNS-bedieners.* (2)
- 5.5 Two of the network devices' IP addresses cannot be used on the Internet. /  
*Twee van die netwerktoestelle se IP-adresse kan nie op die Internet gebruik word nie.*
- 5.5.1 Identify these two devices by writing down their fully qualified domain names. /  
*Identifiseer hierdie twee toestelle deur hul volle gekwalifiseerde domeinnaam neer te skryf.* (4)
- 5.5.2 What protocol can be implemented to enable these devices to gain access to the Internet by means of a single shared IP address? /  
*Watter protokol kan geïmplementeer word om hierdie toestelle in staat te stel om toegang tot die Internet te verkry deur middel van 'n enkele gedeelde IP-adres?* (1)
- 5.5.3 Explain how this protocol functions. /  
*Verduidelik hoe hierdie protokol funksioneer.* (3)
- 5.6 A workstation on the LAN sent a request to the web server to view a web page. /  
*'n Werkstasie op die LAN het 'n versoek aan die webbediener gestuur om na 'n webblad te kyk.*
- 5.6.1 Write down the destination socket that was used for the request. /  
*Skryf die bestemming-sok wat vir die versoek gebruik was neer.* (2)
- 5.6.2 Explain in detail the role that the DNS protocol played during the transmission of the request with regard to local, top level and root name servers. /  
*Verduidelik volledig die rol wat die DNS-protokol gespeel het tydens die oordrag van die versoek met betrekking tot plaaslike, boonste-vlak en hoof naam-bedieners.* (4)
- 5.7 Explain the disadvantages of hosting multiple virtual servers on a single host machine. /  
*Verduidelik die nadele daarvan om verskeie virtuele bedieners op 'n enkele gasheer masjien te huisves.* (4)

5.8 Which virtual network connection type will the virtual servers in the DMZ have to use? /

*Watter virtuele netwerk-konneksietipe sal deur die virtuele bedieners in die DMZ gebruik moet word?*

(1)

5.8.1 Give a reason for your answer in 5.8. /

*Gee 'n rede vir jou antwoord in 5.8.*

(1)

[26]

**End of Paper / Einde van Vraestel**